

OXPLive MCP Server

Secure AI Access to the OXPLive Platform

Prepared for	OrthoKinetix (ORX)
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Project	OXPLive Modernization — Phase B Deliverable
Date	June 2026
Status	Pending Client Approval — Phase F (Build)
Classification	Confidential — OrthoKinetix / Technijian Use Only



Companion document: STP-API-MCP Coverage Matrix (A3 landscape) — all 116 stored procedures, API endpoints, and MCP tool definitions. Available on request.

01 Executive Summary

OrthoKinetix operates the **OXPLive** practice management platform — a mature SQL Server system running the full DME order lifecycle: intake, benefits verification, insurance authorization, shipping, billing, re-billing, and collections across 13 workflow queues.

Technijian proposes to make that data safely available to AI assistants (Claude) through a purpose-built **Model Context Protocol (MCP) server**. The server connects Claude directly to the OXPLive database via a set of 116 curated stored procedures across 20 functional domains — without granting the AI any direct table access.

The governing rule: every AI request follows a single, auditable path — Claude calls a documented MCP tool → the tool executes an approved stored procedure → the stored procedure reads or writes the database. No exceptions. No inline SQL. No table-level permissions for the AI service account.

This document presents the complete scope of what will be built, the 116-procedure coverage matrix, the security and HIPAA controls in place, the delivery plan, and the approval request to begin Phase F (Build).

The full engineering specification — including every stored procedure, its parameters, return schema, API endpoint, and MCP tool definition — is documented in the companion *STP-API-MCP Coverage Matrix*, provided as a separate deliverable.

02 Why Now — The Current-State Gap

Technijian has been the active IT and development partner for OrthoKinetix since approximately 2015. As part of the ongoing **OXPLive Modernization** project (migrating from ColdFusion to .NET 8 / React 18 under Technijian SDLC v6.0), a complete database-call inventory was conducted in early 2026.

What Was Found

The legacy OXPLive ColdFusion frontend makes hundreds of direct database calls across 13 workflow queues. These calls are not governed — they bypass stored procedures in many places, embed business logic in the UI layer, and contain hardcoded credentials. This creates three gaps that block safe AI access today:

- **No single governed access path.** The application reaches the database through a mix of stored procedures, inline SQL, and direct table reads. An AI assistant granted the same access would have uncontrolled write access to production data.
- **No audit trail by design.** Direct database calls leave no application-level record of who accessed what, when, or why — a HIPAA audit requirement.
- **No semantic boundary.** Without a procedure layer, there is no clean way to expose only what the AI should see while hiding raw PHI fields, financial data, and operational tables.

The Solution

The OXPLive Modernization addresses all three gaps by establishing a complete, SP-Only database access pattern. Every frontend interaction — and every AI tool call — must go through a named, documented, version-controlled stored procedure. This is both a modernization best practice and the prerequisite for safe AI integration.

03 Architecture — DB-Direct Design

The OXPLive MCP server uses a **DB-Direct** architecture: Claude connects to the MCP server via Streamable HTTP, and the server executes stored procedures directly against SQL Server via a secure connection pool. There is no intermediate REST API layer for MCP access.



Why DB-Direct

The OXPLive Modernization Phase F will build a .NET 8 REST API, but the stored procedures are being designed now as part of Phase B. DB-Direct lets the MCP server go live the moment the stored procedures are tested — before the full API layer is complete — giving OrthoKinetix early AI access while the broader modernization continues.

Security posture is unchanged from an API-first design. The SQL Server service account (oxplive_mcp_svc) holds only EXECUTE on the oxadmin schema. It cannot read tables directly, run dynamic SQL, or access any object outside the curated procedure set.

Transport — Streamable HTTP

The server implements the **MCP 2025-03-26 Streamable HTTP** transport specification:

- **POST /mcp** — creates or resumes a session; handles all JSON-RPC tool calls
- **GET /mcp** — SSE stream for server-initiated messages (session-keyed)
- **DELETE /mcp** — gracefully closes a session
- **GET /health** — returns { status: "ok", sessions: N }

Sessions are identified by the mcp-session-id header, enabling multiple concurrent Claude sessions against the same server without interference.

04 Domain Coverage Matrix

The table below summarises the complete stored-procedure coverage across all 20 functional domains of the OXPLive system. Every domain corresponds to a distinct area of the OXPLive workflow UI. The full 116-row matrix — with individual SP parameters, return schemas, API endpoint definitions, and MCP tool specifications — is provided in the companion *STP-API-MCP Coverage Matrix* document.

#	Domain	STPs	API Endpoints	MCP Tools	Business Purpose
1	Auth & Session	5	4	2	User authentication, token lifecycle, session management. Most SPs are internal; one tool surfaces current-user context to AI assistants.
2	Patient / Account	18	16	16	Core patient record — demographics, insurance, job creation, medical history, diagnosis updates, status, and cancellation. Largest domain in the system.
3	Insurance Authorization	5	5	5	Auth request creation, line-item management, status tracking, and approval workflow for DME insurance authorizations.
4	Queue & Workflow	7	7	6	13-stage workflow queue (Intake → Billing → Collections). Job transitions, queue-level counts, and routing rules for DME order progression.
5	Billing & Claims	5	5	5	Fee schedule lookup, billing search, claims processing, and remittance management for OrthoKinetix revenue cycle.
6	Scheduling	10	9	9	Appointment creation, calendar search, therapist scheduling, and 48-hour delivery alert monitoring for field operations.
7	Patient Experience	7	6	5	PSAT surveys, post-delivery call-back tracking, satisfaction scores, and patient-outcome follow-up.
8	Collector Actions	5	5	4	Daily collector call logs, AR action tracking, daily-total summaries, and collection queue management.
9	Products & Inventory	10	10	9	Equipment catalog, SKU management, inventory levels, and assignment of equipment to specific patients at delivery.
10	Orders & Purchasing	3	3	3	Purchase-order listing, vendor ordering, and reorder tracking to maintain equipment supply.
11	Documents / Files	4	4	3	Document metadata lookup and HIPAA-compliant signed download URLs for CMN, EDF, and other PHI-attached files.

#	Domain	STPs	API Endpoints	MCP Tools	Business Purpose
12	Directory, Users & Reps	9	8	6	Doctor and facility directory, sales representative management, referring physician lookup, and staff-user directory.
13	Time Records (HR)	9	9	6	Timecard entry, manager approval workflow, and payroll-ready time export for field and office staff.
14	Lookups (Generic)	4	4	2	Shared reference data — insurance payer names, territories, status codes, and all system dropdown values.
15	Config	3	3	2	System-wide configuration and company-settings retrieval for runtime behavior and multi-tenant configuration.
16	Reporting	4	4	4	Billing collections report, PSR (physician summary), variance analysis, and KPI dashboard data.
17	Audit & Activity Logs	4	4	3	HIPAA-compliant audit trail — user access log, data-change history, and compliance event recording.
18	Compliance / Scheduled	2	2	2	Mass job-move operations and compliance batch processing for regulatory workflows.
19	Yes/No Module (PA/CMN/EDF)	1	1	1	Three compliance gates that block queue progression: Prior Authorization status, CMN receipt confirmation, and EDF sign-off.
20	Dashboard	1	1	1	Real-time territory and user activity dashboard — session-aware, role-filtered activity feed.
TOTAL		116	110	94	100% of OXPLive database calls covered

STPs = Stored Procedures. API Endpoints = planned .NET 8 REST endpoints (Phase F). MCP Tools = tools exposed to Claude today via the DB-Direct MCP server. Some STPs are INTERNAL (called by other SPs) and are not exposed as MCP tools.

05 Detailed Capability Reference

Every stored procedure the OXPLive MCP server exposes is listed below, grouped by functional domain. Each row shows whether the operation **reads** data or **writes / updates** it, the API endpoint that will map to this procedure in Phase F, and a plain-English description of what OrthoKinetix staff and AI assistants can do with it.

73

Read Operations

34

Write / Update Operations

9

Internal (not exposed)

Domain 1 — Auth & Session

#	Stored Procedure	R / W	API Endpoint	What It Does
1.1	usp_Auth_Ensure User	INTERNAL	<i>not exposed</i>	INTERNAL — confirms/creates the Azure AD user on first login; called by API middleware only
1.2	usp_Auth_GetCurrentUser	READ	GET /api/me	Returns the logged-in user's name, email, roles, division, and territory to populate the app shell
1.3	usp_Auth_GetIdp Config	INTERNAL	<i>not exposed</i>	INTERNAL — returns MSAL configuration for the SPA; agents use client_credentials instead
1.4	usp_Auth_LogSessionHistory	INTERNAL	<i>not exposed</i>	INTERNAL — records every page navigation for the Admin activity log; called by request pipeline
1.5	usp_Auth_RefreshToken	INTERNAL	<i>not exposed</i>	INTERNAL — silently rotates access/refresh token pair when the SPA token expires; MSAL-managed

Domain 2 — Patient / Account

#	Stored Procedure	R / W	API Endpoint	What It Does
2.1	usp_Patient_Search	READ	GET /api/patients/search	Find patients by name, account number, DOB, SSN-last-4, or phone — powers the global search bar
2.2	usp_Patient_GetDetail	READ	GET /api/patients/{jobId}	Pull the complete patient record: demographics, insurance, medical info, referring doctor, current status, assigned reps
2.3	usp_Patient_Create	WRITE	POST /api/patients	Create a new patient account with demographics, insurance, and medical intake data in a single transaction
2.4	usp_Patient_Update	WRITE	PUT /api/patients/{jobId}	Save edits to an existing patient record; changes are audited before/after
2.5	usp_Patient_GetSnapshot	READ	GET /api/patients/{jobId}/snapshot	Retrieve a complete printable snapshot: all products, billing codes, insurance, clinical notes in one call

#	Stored Procedure	R / W	API Endpoint	What It Does
2.6	usp_Patient_GetCoverage	READ	GET /api/patients/{jobId}/coverage	Get payer name, policy number, group number, coverage type, and current auth numbers
2.7	usp_Patient_UpdateCoverage	WRITE	PUT /api/patients/{jobId}/coverage	Update primary or secondary insurance coverage without touching demographics
2.8	usp_Patient_GetOrigRep	READ	GET /api/patients/{jobId}/orig-rep	Get the original rep assigned at intake — used to preserve commission origination credit
2.9	usp_Patient_SetOrigRep	WRITE	PUT /api/patients/{jobId}/orig-rep	Set or update the originating rep when first assigned during intake
2.10	usp_Patient_ListProducts	READ	GET /api/patients/{jobId}/products	List all product/equipment line items on the account with HCPCS codes, billing status, and auth notes
2.11	usp_Patient_AddProduct	WRITE	POST /api/patients/{jobId}/products	Add a new product line item to the account; validates HCPCS code mapping
2.12	usp_Patient_UpdateProduct	WRITE	PUT /api/patients/{jobId}/products/{id}	Update quantity or notes on a product line item
2.13	usp_Patient_RemoveProduct	WRITE	DELETE /api/patients/{jobId}/products/{id}	Remove a product line item; blocked if a claim has already been submitted for it
2.14	usp_Patient_ListNotes	READ	GET /api/patients/{jobId}/notes	Read all free-text notes on the account in chronological order
2.15	usp_Patient_AddNote	WRITE	POST /api/patients/{jobId}/notes	Append a note to the account; notes are immutable once created
2.16	usp_Patient_GetAgreement	READ	GET /api/patients/{jobId}/agreement	Check whether the patient has accepted the service agreement and on what date
2.17	usp_Medical_UpdateDiagnosis	WRITE	PUT /api/patients/{jobId}/medical/diagnosis	Update ICD-10 diagnosis codes, referring doctor, body part, and procedure code
2.18	usp_Patient_Cancel	WRITE	POST /api/patients/{jobId}/cancel	Cancel the account with a required reason code and notes; writes a final status history entry

Domain 3 — Insurance Authorization

#	Stored Procedure	R / W	API Endpoint	What It Does
3.1	usp_Auth_ListByPatient	READ	GET /api/patients/{jobId}/authorizations	List all insurance authorizations on the account: auth numbers, dates,

#	Stored Procedure	R / W	API Endpoint	What It Does
				payer, status, and linked products
3.2	usp_Auth_Create	WRITE	POST /api/patients/{jobId}/authorizations	Create a new insurance authorization record with auth number, dates, payer, and notes
3.3	usp_Auth_Update	WRITE	PUT /api/authorizations/{authId}	Update an existing authorization — correct dates, extend, or change status
3.4	usp_Auth_AddItem	WRITE	POST /api/authorizations/{authId}/items	Link a product line item to an authorization; required for billing to confirm each line is covered
3.5	usp_Auth_RemoveItem	WRITE	DELETE /api/authorizations/{authId}/items/{id}	Remove a product from an authorization; blocked if the claim has been submitted

Domain 4 — Queue & Workflow

#	Stored Procedure	R / W	API Endpoint	What It Does
4.1	usp_Queue_List	READ	GET /api/queues	List all 13 workflow queues visible to the current user with current job counts
4.2	usp_Queue_GetJobs	READ	GET /api/queues/{queueId}/jobs	Get all patient accounts currently in a specific queue — paged, with name, status age, and rep
4.3	usp_Queue_TransitionJob	WRITE	POST /api/queues/transition	Move a patient account from one queue to another; validates the transition is permitted, writes status history
4.4	usp_Queue_GetStatusHistory	READ	GET /api/patients/{jobId}/status-history	Get the complete queue-transition history for an account — every move ever made with user and timestamp
4.5	usp_Queue_GetActions	READ	GET /api/queues/{queueId}/actions	List the permitted moves out of a queue — used to render the correct action buttons
4.6	usp_Queue_GetBillingStatus	READ	GET /api/patients/{jobId}/billing-status	Get the billing status badge for an account: Unbilled / Partially Billed / Fully Billed / Cash
4.7	usp_Queue_GetCounts	READ	GET /api/queues/counts	Get the count of accounts in each of the 13 queues in one call — powers the sidebar badges and dashboard

Domain 5 — Billing & Claims

#	Stored Procedure	R / W	API Endpoint	What It Does
5.1	usp_Billing_ListEntities	READ	GET /api/billing/entities	List all billing entities (insurance companies, payers) — used in billing forms and patient intake

#	Stored Procedure	R / W	API Endpoint	What It Does
5.2	usp_Billing_Search	READ	GET /api/billing/search	Search billable records by account number, payer, or status — returns line items with charge and collection data
5.3	usp_Billing_GetFeeSchedules	READ	GET /api/billing/fee-schedules	Get contracted fee schedule amounts for a HCPCS code and billing entity — used to calculate expected reimbursement
5.4	usp_Billing_GetProductValuation	READ	GET /api/billing/valuations/{productId}	Get pricing and rental term options (capped rental, rent-to-own, purchase) for a product
5.5	usp_Billing_GetReBillData	READ	GET /api/patients/{jobId}/rebill	Get previously billed line items flagged for re-billing due to denial, partial payment, or correction

Domain 6 — Scheduling

#	Stored Procedure	R / W	API Endpoint	What It Does
6.1	usp_Schedule_GetByRange	READ	GET /api/schedules	List scheduled appointments filtered by date range, territory, or rep — powers the calendar view
6.2	usp_Schedule_Create	WRITE	POST /api/schedules	Create a new appointment for a patient — type, date/time, assigned rep, and notes
6.3	usp_Schedule_Update	WRITE	PUT /api/schedules/{scheduleId}	Reschedule an appointment or update its type or notes
6.4	usp_Schedule_Delete	WRITE	DELETE /api/schedules/{scheduleId}	Cancel and remove a scheduled appointment; also removes the Exchange calendar sync record
6.5	usp_Schedule_GetUnscheduled	READ	GET /api/schedules/unscheduled	Find accounts in the scheduling queue with no upcoming appointment — lets dispatchers fill the calendar
6.6	usp_Schedule_GetTypes	READ	GET /api/lookups/schedule-types	Get appointment type dropdown (delivery, follow-up, fitting, re-evaluation, etc.)
6.7	usp_Schedule_GetExchangeSync	READ	GET /api/schedules/{scheduleId}/exchange-sync	Get the Outlook calendar sync status for an appointment — shows whether it has been pushed to Exchange
6.8	usp_Schedule_SetExchangeSync	WRITE	PUT /api/schedules/{scheduleId}/exchange-sync	Link an appointment to an Outlook calendar event after it has been pushed to Exchange
6.9	usp_Schedule_DeleteExchangeSync	WRITE	DELETE /api/schedules/{scheduleId}/exchange-sync	Remove the Exchange calendar link when an appointment is cancelled or the sync breaks

#	Stored Procedure	R / W	API Endpoint	What It Does
6.10	usp_Schedule r_Get48HourA lerts	READ	GET /api/schedules/alerts	Get all deliveries and follow-ups due within the next 48 hours for the user's territory — same-day alert panel

Domain 7 — Patient Experience (PSR)

#	Stored Procedure	R / W	API Endpoint	What It Does
7.1	usp_PatientExp erience_List	READ	GET /api/patients/{jobId}/experiences	List all patient contacts on the account — calls, inquiries, complaints — in chronological order
7.2	usp_PatientExp erience_Create	WRITE	POST /api/patients/{jobId}/experiences	Log a patient contact: call attempt, inquiry, complaint, or scheduling contact with outcome
7.3	usp_PatientExp erience_Update	WRITE	PUT /api/patient-experiences/{peId}	Update the status, follow-up date, or notes on a patient contact record (close out or escalate)
7.4	usp_PatientExp erience_GetTyp es	READ	GET /api/lookups/patient-experience- types	Get contact type dropdown (inquiry, scheduling, complaint, satisfaction, etc.)
7.5	usp_PatientExp erience_GetSta tuses	READ	GET /api/lookups/patient-experience- statuses	Get contact status dropdown (open, resolved, escalated, etc.)
7.6	usp_PatientExp erience_GetCom plaintTypes	READ	GET /api/lookups/complaint-types	Get complaint type dropdown — used when logging a formal patient complaint
7.7	usp_PatientExp erience_GetTem plates	READ	GET /api/lookups/patient-experience- templates	Get message templates that pre-populate contact notes for common contact types

Domain 8 — Collector Actions

#	Stored Procedure	R / W	API Endpoint	What It Does
8.1	usp_CollectorAc tion_List	READ	GET /api/collector-actions	List collector actions for an account or by rep — calls made, voicemails, payment arrangements
8.2	usp_CollectorAc tion_Create	WRITE	POST /api/collector-actions	Log a new collection activity: call made, voicemail left, payment plan set, referral to agency
8.3	usp_CollectorAc tion_Update	WRITE	PUT /api/collector- actions/{actionId}	Update a collector action — revise the follow-up date or add resolution notes
8.4	usp_CollectorAc tion_GetReasonC odes	READ	GET /api/lookups/collector- action-reasons	Get reason code dropdown (called/no answer, payment plan set, referred to agency, etc.)

#	Stored Procedure	R / W	API Endpoint	What It Does
8.5	usp_CollectorAction_GetDailyTotals	READ	GET /api/collector-actions/totals	Get aggregated collector performance stats — actions, follow-ups completed, accounts touched — by user and date range

Domain 9 — Products & Inventory

#	Stored Procedure	R / W	API Endpoint	What It Does
9.1	usp_Product_List	READ	GET /api/products	Browse the product catalog filtered by category or active status
9.2	usp_Product_Create	WRITE	POST /api/products	Add a new product to the catalog with pricing and vendor info (admin only)
9.3	usp_Product_Update	WRITE	PUT /api/products/{productId}	Update product name, part number, pricing, or active status (admin only)
9.4	usp_Product_GetBillingCodes	READ	GET /api/products/{productId}/billing-codes	Get HCPCS billing code mappings for a product — used in billing validation
9.5	usp_Product_GetBodyParts	READ	GET /api/lookups/body-parts	Get body part dropdown (Left Knee, Right Ankle, etc.) for the medical intake section
9.6	usp_Product_GetCategories	READ	GET /api/lookups/product-categories	Get product category dropdown (AFO, KAFO, TLSO, etc.)
9.7	usp_Product_GetManufacturers	READ	GET /api/lookups/manufacturers	Get vendor/manufacturer dropdown for product assignment
9.8	usp_Inventory_ListLocations	READ	GET /api/inventory/locations	List warehouse and storage locations by site — where stock is physically held
9.9	usp_Inventory_ListSites	READ	GET /api/inventory/sites	List inventory sites (warehouses and offices) — parent lookup for inventory locations
9.10	usp_Inventory_Assign	WRITE	PUT /api/inventory/{equipmentId}/assign	Assign a specific piece of serialized equipment to a patient — records serial number, model, and assignment date for stock-and-bill compliance

Domain 10 — Orders & Purchasing

#	Stored Procedure	R / W	API Endpoint	What It Does
10.1	usp_Order_ListByPatient	READ	GET /api/patients/{jobId}/orders	List all orders and purchase orders linked to the patient account with status and item count
10.2	usp_Order_GetDetail	READ	GET /api/orders/{orderId}	Get full order detail including all line items with SKU, description, quantity, and price

#	Stored Procedure	R / W	API Endpoint	What It Does
10.3	usp_Purchasing_GetByPatient	READ	GET /api/patients/{jobId}/purchasing	Get the sales order number, PO number, and invoice number for the account's purchasing record

Domain 11 — Documents / Files

#	Stored Procedure	R / W	API Endpoint	What It Does
11.1	usp_Document_ListByPatient	READ	GET /api/patients/{jobId}/documents	List all documents attached to the account: type, filename, upload date, and uploader
11.2	usp_Document_GetMeta	INTERNAL	<i>not exposed</i>	INTERNAL — retrieves blob path for signed URL generation; raw path never returned to client (HIPAA)
11.3	usp_Document_CreateRecord	WRITE	POST /api/patients/{jobId}/documents	Attach an uploaded file to the patient account; blob is stored in Azure, metadata recorded in the database
11.4	usp_Document_GetViewUrl	READ	GET /api/documents/{documentId}/view	Get a 5-minute signed URL to view a patient document; access is logged to the HIPAA audit trail automatically

Domain 12 — Directory, Users & Reps

#	Stored Procedure	R / W	API Endpoint	What It Does
12.1	usp_Directory_GetList	READ	GET /api/directory	Browse the staff directory filtered by division or territory — used for rep assignment pickers
12.2	usp_Directory_GetReps	READ	GET /api/lookups/reps	Get the sales rep list, optionally filtered by territory — populates rep assignment dropdowns
12.3	usp_Directory_GetTerritories	READ	GET /api/lookups/territories	Get the territory lookup (regions and territories) — used in scheduling, intake, and queue filtering
12.4	usp_Directory_GetDivisions	READ	GET /api/lookups/divisions	Get the division/business-unit lookup
12.5	usp_Directory_GetDoctors	READ	GET /api/doctors	Browse the physician directory filtered by specialty — used on the referring doctor picker
12.6	usp_Directory_GetPractitioners	READ	GET /api/practitioners	List clinical practitioners (PTs, OTs, RNs) — used on clinical assignment pickers
12.7	usp_User_ListAll	READ	GET /api/admin/users	List all user accounts with roles, division, and territory (admin only — User Management)

#	Stored Procedure	R / W	API Endpoint	What It Does
				screen)
12.8	usp_User_GetSecurity	READ	GET /api/admin/users/{userId}/security	Get role assignments and territory access for a specific user (admin only)
12.9	usp_User_GetRoleTypes	READ	GET /api/admin/roles	Get all defined role types (Admin, Billing, Clinical, etc.) — used in user role assignment

Domain 13 — Time Records (HR)

#	Stored Procedure	R / W	API Endpoint	What It Does
13.1	usp_TimeRecord_List	READ	GET /api/time-records	List time record submissions filtered by employee, pay period, or approval status
13.2	usp_TimeRecord_GetDetail	READ	GET /api/time-records/{timeRecordId}	Get a specific time record with all hourly or salaried line items
13.3	usp_TimeRecord_Create	WRITE	POST /api/time-records	Create a new time record header for a pay period
13.4	usp_TimeRecord_Update	WRITE	PUT /api/time-records/{timeRecordId}	Submit, approve, or reject a time record — workflow status update
13.5	usp_TimeRecord_GetSalariedData	READ	GET /api/time-records/{id}/salaried-data	Get daily salaried entries and PTO allocations (included in GetDetail response)
13.6	usp_TimeRecord_GetHourlyData	READ	GET /api/time-records/{id}/hourly-data	Get hourly shift entries with start/end times and OT hours (included in GetDetail response)
13.7	usp_TimeRecord_GetPTOTypes	READ	GET /api/lookups/pto-types	Get PTO type dropdown (Vacation, Sick, Personal, Holiday, Bereavement)
13.8	usp_TimeRecord_GetPaymentCalendar	READ	GET /api/time-records/payment-calendar	Get pay period dates for a given year — used to validate week ranges on timecard entry
13.9	usp_TimeRecord_GetHolidayCalendar	READ	GET /api/time-records/holiday-calendar	Get company holidays for a year — used in timecard calculations for days that count as worked

Domain 14 — Lookups (Generic)

#	Stored Procedure	R / W	API Endpoint	What It Does
14.1	usp_Lookup_GetValues	READ	GET /api/lookups/{tableName}	Get dropdown values for any lookup table by name — cancellation reasons, document types, regions, etc. One SP covers all reference lists
14.2	usp_Lookup_Create	WRITE	POST /api/admin/lookups/{tableName}	Add a new value to a lookup table (admin only); change is audit-logged

#	Stored Procedure	R / W	API Endpoint	What It Does
14.3	usp_Lookup_Update	WRITE	PUT /api/admin/lookups/{tableName}/{id}	Update a lookup table value (admin only); change is audit-logged
14.4	usp_Lookup_Delete	INTERNAL	not exposed	INTERNAL — soft-deletes a lookup value; blocked if value is in use; admin UI action only, not an MCP tool

Domain 15 — Config

#	Stored Procedure	R / W	API Endpoint	What It Does
15.1	usp_Config_GetIdp	INTERNAL	not exposed	INTERNAL — returns MSAL configuration for the SPA; agents use client_credentials, not IDP config
15.2	usp_Config_GetAll	READ	GET /api/config	Get all system configuration values: session timeout, max file size, feature flags, runtime settings
15.3	usp_Config_GetCompany	READ	GET /api/config/company	Get company name, address, and branding for display in reports and print views

Domain 16 — Reporting

#	Stored Procedure	R / W	API Endpoint	What It Does
16.1	usp_Report_BillingCollections	READ	GET /api/reports/billing-collections	Billing and collections report: outstanding balances, collection performance by payer, aging buckets
16.2	usp_Report_DepartmentDeliverables	READ	GET /api/reports/department-deliverables	Department productivity report: deliverables completed, time logged, productivity score by division
16.3	usp_Report_PSR	READ	GET /api/reports/psr	Patient Satisfaction (PSR) report: contact outcomes, complaint resolution, rep performance by date range
16.4	usp_Report_Variance	READ	GET /api/reports/variance	Pricing variance report: billed amounts vs contracted rates — surfaces underbilling, overbilling, and margin issues

Domain 17 — Audit & Activity Logs

#	Stored Procedure	R / W	API Endpoint	What It Does
17.1	usp_AuditLog_Insert	INTERNAL	not exposed	INTERNAL — writes HIPAA audit events (PHI access, data change, security); called by API middleware only
17.2	usp_AuditLog_GetList	READ	GET /api/admin/audit-logs	Search the HIPAA audit log: PHI access events, data changes, security events (admin/auditor only)
17.3	usp_ActivityLog_Insert	INTERNAL	not exposed	INTERNAL — records page views, logins, and logouts; called by request pipeline middleware

#	Stored Procedure	R / W	API Endpoint	What It Does
				only
17.4	usp_ActivityLog_GetList	READ	GET /api/admin/activity-logs	Search the user activity log: login, logout, and page-view events by user and date range (admin only)

Domain 18 — Compliance / Scheduled

#	Stored Procedure	R / W	API Endpoint	What It Does
18.1	usp_Compliance_GetPendingJobs	READ	GET /api/admin/compliance/pending	List patient accounts awaiting compliance review that are overdue — used for the daily compliance sweep
18.2	usp_Compliance_MassMoveJobs	WRITE	POST /api/admin/compliance/mass-move	Bulk-move accounts that have exceeded the compliance hold period to the next queue; returns count of accounts moved

Domain 19 — Yes/No Module (PA/CMN/EDF)

#	Stored Procedure	R / W	API Endpoint	What It Does
19.1	usp_YesNo_Update	WRITE	PUT /api/patients/{jobId}/yesno	Update the three compliance gates in one call: PA authorization status, CMN receipt confirmation, and EDF sign-off. PA Denied blocks the account from advancing to Billing

Domain 20 — Dashboard

#	Stored Procedure	R / W	API Endpoint	What It Does
20.1	usp_Dashboard_GetActivity	READ	GET /api/dashboard/activity	Get six headline dashboard KPIs in one call: new intakes today, queue transitions, scheduled deliveries, documents uploaded, open collector follow-ups, overdue compliance. Territory-filtered for non-admin users

06 Security, HIPAA & Governance

OXPLive contains protected health information — patient demographics, insurance details, DME order data, and EDI 835/837 claims. Security controls are designed into every layer of this architecture, not added afterward. The following table summarises each control.

Control	Implementation
Service Account Isolation	The <code>oxplive_mcp_svc</code> SQL Server login holds only <code>GRANT EXECUTE ON SCHEMA::oxadmin</code> . No table-level <code>SELECT</code> , <code>INSERT</code> , <code>UPDATE</code> , or <code>DELETE</code> permissions are granted. No dynamic SQL is possible.
SP-Only Access Pattern	Every AI request resolves to a named, version-controlled stored procedure. Inline SQL from AI-generated queries is architecturally impossible.
Audit Logging	Domain 17 exposes audit-log procedures. HIPAA-sensitive SPs (<code>usp_Document_GetViewUrl</code> , all patient-record writers) trigger server-side audit entries recording user, action, timestamp, and affected record ID.
PHI Field Masking	SSN and other PHI identifiers are masked at the stored-procedure level before they reach the MCP response payload. Patient-data fields require explicit role clearance (enforced in the procedure's parameter validation).
Document URLs	File access (<code>usp_Document_GetViewUrl</code>) returns HIPAA-compliant signed URLs with short-lived expiry. No raw file paths or permanent links are returned.
Transport Security	Streamable HTTP runs over HTTPS in production. An optional <code>MCP_API_KEY</code> header provides HTTP-level bearer auth. Network access to the MCP server port is restricted by firewall to authorised client IP ranges.
Session Isolation	Each Claude session receives a unique <code>mcp-session-id</code> . Sessions are server-memory-scoped; one session cannot access another session's state or results.
Compliance Domains	Domain 19 (Yes/No Module) enforces the three compliance gates — PA Status, CMN receipt, and EDF sign-off — that must be satisfied before a job can progress in the queue. Domain 18 handles mass-move and scheduled compliance batch operations.

This design supports OrthoKinetix's HIPAA compliance posture. It does not replace the organisation's existing policies, Business Associate Agreements, or staff training, which remain in place.

07 What Gets Built

MCP Server — Node.js Application

The OXPLive MCP server is a **Node.js / TypeScript** application built on the official `@modelcontextprotocol/sdk` package. It uses the `mssql` library for SQL Server connectivity and Express for the Streamable HTTP transport. Key components:

- **DB Client** — singleton `mssql` connection pool; `executeSP()`, `querySP()`, and `querySPSingle()` helpers that auto-prefix `oxadmin.` to every procedure name
- **20 Domain Tool Files** — one TypeScript file per domain, each exporting its MCP tool definitions and request handlers
- **Server Core** — `createOxpliveMcpServer()` registers all 94 tools and dispatches calls to the matching domain handler
- **Streamable HTTP Transport** — Express app with session map; supports multiple concurrent Claude sessions

SQL Server Service Account

A dedicated, least-privilege SQL Server login is created specifically for the MCP server:

```
CREATE LOGIN oxplive_mcp_svc WITH PASSWORD = '<vault-managed password>';
CREATE USER oxplive_mcp_svc FOR LOGIN oxplive_mcp_svc;
GRANT EXECUTE ON SCHEMA::oxadmin TO oxplive_mcp_svc;
```

No further permissions are required. The service account cannot read, write, or alter any table, view, or function directly.

Configuration & Deployment

All connection strings and secrets are managed via environment variables (never committed to source control). The server ships with a `.env.example` file and a `config/claude-code.mcp.json` snippet for immediate Claude Code integration.

What Is NOT in Scope

- The .NET 8 REST API layer (Phase F — planned, not yet approved)
- An admin portal or management dashboard
- Changes to the existing OXPLive ColdFusion application
- Changes to the OrthoXpressDB schema or existing stored procedures

08 Delivery Plan

The MCP server delivery is structured in three phases. Each phase ends with a clear, demonstrable result that OrthoKinetix can test before the next phase begins.

Phase	Deliverable	Scope	Exit Criteria
Phase 1	Stored Procedure Foundation	All 116 stored procedures authored, reviewed, and deployed to the OXPLive test database. Includes read-only procedures for all 20 domains plus write procedures for core entities.	All SPs pass unit tests against test DB; naming convention compliant; HIPAA-sensitive fields verified
Phase 2	MCP Server — Build & Test	Node.js MCP server with all 94 tools, DB-Direct wiring, Streamable HTTP transport, service account provisioning, and Claude Code integration.	Claude can invoke all 94 tools against the test database; audit log confirms all calls recorded
Phase 3	Production Deployment	Production environment setup, credential vault integration, firewall rules, HTTPS, staff training session, and final sign-off from OrthoKinetix IT.	Server running in production; OrthoKinetix team can connect Claude and execute representative queries

Detailed timeline and effort estimates are confirmed during the kick-off session and provided in a companion Statement of Work upon approval of this document.

09 What We Need From You

Approval of this document authorises Technijian to begin Phase 1. A short kick-off session is scheduled immediately afterward to confirm the items below. None of these decisions block approval — they shape the detailed plan.

- **Domain priority order.** Which of the 20 domains should be built first? (Suggested default: Domains 1–5 — Auth, Patient, Insurance Auth, Queue, Billing.)
- **Test database access.** Confirmation that Technijian may continue working against the existing test/UAT environment for Phase 1 and Phase 2.
- **Service account provisioning.** Authorisation to create the `oxplive_mcp_svc` login and grant it EXECUTE on the `oxadmin` schema.
- **Credential vault.** Preferred secret-management approach (Azure Key Vault, Windows Credential Manager, or Technijian-managed vault) for the service account password and API key.
- **Hosting location.** Where the MCP server will run — on-premises IIS, Windows service, Docker container, or Technijian-hosted.
- **Internal users.** Who at OrthoKinetix will use Claude with the MCP tools, and at what initial permission level (read-only vs. read/write access to write-capable domains).

10 Approval & Sign-Off

By signing below, OrthoKinetix authorises Technijian to proceed with the OXPLive MCP Server project as described in this document, beginning with Phase 1 (Stored Procedure Foundation). This approval covers the technical approach, domain scope, and phased delivery plan.

Specific timeline, effort hours, and fees are confirmed in a companion Statement of Work, which will be issued within five (5) business days of receiving this signed approval.

OrthoKinetix

Authorised on behalf of OrthoKinetix / OrthoXpress

Signature Date

Name (print) Title

Technijian

Prepared and submitted by Technijian Inc.

Rohit Jain, Technijian June 2026

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Questions? Contact Technijian at 949.379.8499 (main) or rjain@technijian.com. The full engineering specification, STP-API-MCP Coverage Matrix, and Statement of Work are available on request.

A Appendix — Full Domain Reference

A.1 Domain Summary (authoritative counts)

#	Domain	STPs	API Endpoints	MCP Tools
1	Auth & Session	5	4	2
2	Patient / Account	18	16	16
3	Insurance Authorization	5	5	5
4	Queue & Workflow	7	7	6
5	Billing & Claims	5	5	5
6	Scheduling	10	9	9
7	Patient Experience	7	6	5
8	Collector Actions	5	5	4
9	Products & Inventory	10	10	9
10	Orders & Purchasing	3	3	3
11	Documents / Files	4	4	3
12	Directory, Users & Reps	9	8	6
13	Time Records (HR)	9	9	6
14	Lookups (Generic)	4	4	2
15	Config	3	3	2
16	Reporting	4	4	4
17	Audit & Activity Logs	4	4	3
18	Compliance / Scheduled	2	2	2
19	Yes/No Module (PA/CMN/EDF)	1	1	1
20	Dashboard	1	1	1
TOTAL		116	110	94

A.2 The 13 Workflow Queues

The OXPLive platform routes DME orders through 13 named queues. Domain 4 (Queue & Workflow) provides the stored procedures that govern job transitions between queues.

#	Queue	Purpose
1	Intake	New DME orders received; initial eligibility check
2	Benefits	Insurance benefit verification in progress

#	Queue	Purpose
3	Auth	Prior authorisation requested or under review
4	Pending	Awaiting patient or clinical information
5	Shipping	Order approved; equipment dispatch in progress
6	Billing	Delivered; claim submitted to payer
7	Re-Bill	Claim denied or adjusted; re-submission in progress
8	EDF	Equipment Delivery Form outstanding — compliance hold
9	Collections	Overdue balance; active collector follow-up
10	Yes/No	PA Status / CMN / EDF compliance gates pending
11	Inactive	Order on hold; no active work
12	Cancelled	Order cancelled; record retained for audit
13	Collector Actions	Collector daily-action tracking and totals

A.3 Companion Documents

- **STP-API-MCP Coverage Matrix** (A3 landscape PDF) — All 116 stored procedures with full parameter lists, return schemas, API endpoint mappings, and MCP tool definitions. Provided as a separate deliverable.
- **Statement of Work** — Issued within 5 business days of approval. Covers phased timeline, effort hours, and fees.